

Spire AI-S2S Dataset Variable Directory

A summary of primary zarr fields, surface/upper air variables, and ensemble derivatives

Variable Group	Variables	Units	Ensemble Derivatives	Description
Surface Temperature	2m Air Temperature	K	Mean, StdDev, Percentiles, Probabilities Anomalies (Min / Max only)	Air temperature at 2 metres above the surface over the 24-hour forecast period.
	2m Max Temperature	K		
	2m Min Temperature	K		
Upper Air Variables	Temperature	K	Mean, StdDev Anomalies (Temp & Height only)	Variables provided at standard isobaric pressure levels.
	Geopotential Height	m		
	Specific Humidity	kg kg ⁻¹		
Precipitation	Accumulated Precipitation	kg m ⁻²	Mean, StdDev, Percentiles, Probabilities, Anomalies	Total accumulated precipitation over the 24-hour forecast period.
Radiation & Fluxes	Downwelling Shortwave Flux	W m ⁻²	Mean, StdDev Percentiles, Probabilities, Anomalies (Shortwave only)	Surface solar radiation and net upward longwave radiation at top of atmosphere.
	Top Thermal Radiation	W m ⁻²		
Surface Wind & Pressure	10m / 100m Wind Speed	m s ⁻¹	Mean, StdDev, Percentiles, Probabilities Anomalies (Speed & MSLP only)	Wind speed, U/V components at 10m/100m and atmospheric pressure at sea level.
	10m / 100m Wind Components	m s ⁻¹		
	Mean Sea Level Pressure	Pa		
Weather Regimes	CONUS Regimes	Probability	Probability	Probability of large-scale circulation regimes over CONUS and Europe-Atlantic.
	EUATL Regimes	Probability		